

NeoRepro 核心成果 | Expert Brief

公共 MHC-I 新抗原预测工具的泄漏感知复现与患者级评估 | Leakage-aware reproduction and patient-level evaluation

一句话结论

最重要的发现不是简单宣布“谁最好”，而是证明：常用外部基准可能已进入模型训练集；严格排除泄漏后，模型差异仍存在，但绝对预测能力有限。

BOTTOM LINE

The key result is not a simplistic winner. A widely used external benchmark had entered training data; after strict leakage exclusion, model differences persisted, but absolute predictive performance remained modest.

520 / 520

TESLA 记录与 PRIME2 训练集精确重叠

TESLA records exactly overlapped PRIME2 training data; they were withdrawn as external validation evidence.

17,475

泄漏过滤后的 IMPROVE pMHC 记录

Leakage-filtered pMHC records: 465 positives, 70 patients, 3 cohorts, one common evaluation set.

+0.051

PRIME 相对 BigMHC 的 AUROC 差值

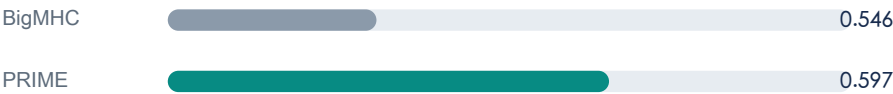
Patient-bootstrap 95% CI: 0.008 to 0.092. Direction robust across all reported sensitivity analyses.

主要结果 | PRIMARY RESULTS

common support / same task

总体区分度 | Pooled AUROC

higher is better



患者级检索 | Mean pMHC Recall@20

60 positive-bearing patients



稳健性 | Robustness

Exact-peptide, Hamming-1, 9-10mer, patient-peptide and within-HLA normalization all preserved the direction.

实质价值 | WHY IT MATTERS

- 阻止错误外部验证
Prevents training-overlapped data from being presented as independent evidence.
- 把比较变成可审计流程
Provides pinned tools, common support, provenance, failure evidence and byte-stable reproduction.
- 给出可信而克制的模型差异
PRIME ranked better here, but low AP and patient heterogeneity argue against universal superiority.

证据边界 | SCOPE

仅评估 presentation 预筛选候选的 pMHC multimer 可检测 T-cell recognition。
Not natural processing, tumor presentation, killing or clinical benefit.

复现验证 | REPRODUCIBILITY

Clean checkout: 27/27 tests passed; 24 independent metric checks; max error 1.1e-16; tracked outputs byte-stable.

主要来源 | KEY SOURCES

IMPROVE (2024) · PRIME2 (2023) · BigMHC (2023) · TESLA (2020)